

Mortality risk due to non-optimum temperatures by cause of death, age, and sex: a multi-country time-series study



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Summary

Background Non-optimum temperatures have been linked to increased mortality, but the cause-, age-, and sex-specific impacts remain largely unclear. This study investigates the temperature–mortality relationships for nine causes of death across multiple countries/territories and explores subgroup differences by sex and age.

Methods We analysed the non-linear and lagged associations between temperature and mortality in 1117 locations from ten countries or territories (Australia, Brazil, Canada, Chile, Mexico, New Zealand, Philippines, South Korea, Taiwan, and Thailand) covering country-specific periods within 2000–2019, using a two-stage time-series design. In the first stage, a quasi-Poisson generalised linear regression with a distributed lag non-linear model was fitted to estimate the location-specific mortality risk associated with temperature. We then applied a meta-regression model in the second stage to synthesise the associations for cause-specific mortality. Stratified analyses were conducted by sex and age group, and the attributable fraction (AF) of mortality was subsequently estimated.

Findings We identified three distinct exposure-response patterns: an inverse J-shaped curve with higher risks from extreme cold for most causes, a U-shaped curve for all-cause and respiratory mortality, and a J-shaped curve for injury and external causes with greater risks at extreme heat. Significant differences in temperature-related mortality risks were observed across age and sex groups, with the direction and magnitude of these differences

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varying by cause of death. We estimated that 2.03 million deaths were attributable to non-optimum temperatures during the study period, corresponding to 4.38% (95% CI: 2.02, 6.59) of all-cause mortality. The highest AFs were observed for mental disorders (6.53%), nervous (6.40%), and cardiovascular causes (5.71%). For most causes of death, cold temperatures accounted for the largest proportion of mortality. However, for deaths related to infectious as well as injury and external causes, heat exposure contributed to the majority of the mortality.

Interpretation Our findings demonstrated substantial variations in temperature-related mortality by cause of death, sex, and age. This analysis highlights the need for tailored public health strategies that address the unique vulnerabilities of specific demographic groups across different causes of death, with targeted interventions to mitigate temperature-related health risks. Future work should focus on improving estimation in data-sparse subgroups and developing cause-, age-, and sex-specific projections of temperature-related health risks under future climate scenarios.

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Keywords: Temperatures; Mortality; Age; Sex; Multi-country study

Research in context

Evidence before this study

We searched PubMed, Web of Science, and Embase from database inception to May 16, 2025 for English-language studies examining the association between non-optimum temperatures and mortality, using search terms “temperature”, “heat”, “cold”, “climate change”, and “mortality”. Previous research has extensively examined the impact of non-optimum temperatures on all-cause and cardiorespiratory mortality. However, mortality risks from other causes related to non-optimum temperatures remain largely unknown. Most studies assessing causes of death beyond cardiorespiratory outcomes were based on analyses from single countries, limiting the generalisability of their findings. Only one global study has examined cause-specific mortality related to non-optimum temperatures. Research gaps persist: lagged effects were not accounted for; key confounders such as seasonality and long-term trends were not adequately adjusted for; and age- and sex-specific mortality risks across different causes of death were not explored. Therefore, previous estimates of temperature-related mortality burden might have been biased, and demographic-specific risks remain unclear.

Added value of this study

To our knowledge, this is the largest multi-country study to date to characterise cause-, age-, and sex-specific mortality risks associated with non-optimum temperatures. We identified three distinct exposure-response patterns: an inverse J-shaped curve with higher risks from extreme cold for most causes, a U-shaped curve for all-cause and respiratory mortality, and a J-shaped curve for injury and

external causes with greater risks at extreme heat. The highest attributable fractions of mortality were observed for mental disorders (6.53%), nervous (6.40%), and cardiovascular causes (5.71%). For most causes of death, cold temperatures contributed the largest share of mortality burden. In contrast, heat exposure accounted for the majority of deaths related to infectious diseases and injury or external causes. Significant differences in temperature-related mortality risks were observed by age and sex. For example, individuals aged ≥60 years faced higher risks from extreme cold for nervous system diseases and neoplasms, whereas those aged 0–59 years experienced significantly higher risks during extreme heat for injury and external causes of death.

Implications of all the available evidence

Understanding the distinct temperature-mortality relationships across various causes of death can help guide targeted interventions. Cold temperatures accounted for the largest mortality burden, emphasising the need for increased caution during cold periods. Conversely, for infectious diseases and injury/external causes, greater attention is required during heat events. The observed variation in age- and sex-specific mortality risks across different causes highlights the importance of tailored public health strategies addressing the unique vulnerabilities of specific demographic groups. Future work should improve estimation in data-sparse subgroups and develop cause-, age-, and sex-specific projections of temperature-related health risks under future climate scenarios.

Introduction

Non-optimum temperatures have been identified as a leading risk factor for global disease burden, contributing to approximately five million deaths annually from 2000 to 2019.^{1,2} These effects are thought to occur primarily through acute physiological stress that triggers death among individuals with underlying chronic conditions.³ While the all-cause and cardiorespiratory mortality risks associated with temperature have been extensively investigated,^{4,5} mortality risks from other causes remain largely unknown, with critical gaps in evidence.⁶ For example, despite mounting evidence that temperature-induced physiological responses (e.g., inflammation and oxidative stress) may exacerbate cancer progression and prognosis, research on temperature-related mortality from neoplastic diseases remains limited.⁷ Similarly, cold-related genitourinary mortality is infrequently investigated, even though cold-induced dehydration has been linked to rhabdomyolysis and acute kidney injury.⁸

Studies have investigated the impacts of temperature on single cause of death, primarily from analyses of a single country (e.g., Japan, China, Germany).^{8–13} However, regional variations in climate, socioeconomic factors, and adaptive capacity limit the generalisability of these findings. Moreover, methodological heterogeneity, particularly in temperature exposure metrics (e.g., daily mean temperature, apparent temperature, wet-bulb globe temperature), study designs, and statistical approaches, further complicates cross-study comparisons and synthesis of evidence.

To date, only one global study has examined cause-specific mortality related to non-optimum temperatures; however, there remain specific research gaps. First, that study did not account for lagged effects, despite substantial evidence that cold-related mortality can persist for up to 21 days.¹⁴ Second, it did not adequately adjust for key confounders such as seasonality and long-term trends, which may have introduced bias into the results.¹⁵ Third, the study did not explore age- and sex-specific mortality risks across different causes of death, despite evidence suggesting that susceptibility to temperature varies by demographic groups.^{16–18} Addressing these knowledge gaps is essential for improving the accuracy and reliability of cause-specific mortality estimates.¹⁵

Taking advantage of data from 1117 locations across ten countries or territories with diverse climates and socio-economic backgrounds, we aimed to characterise cause-, age-, and sex-specific temperature–mortality relationships using a standardised analytical framework and advanced epidemiological methods. We further assessed the cause-specific mortality burden attributable to non-optimum temperatures.

Methods

Data collection

Mortality data

We obtained daily individual death records from 1117 locations across ten countries or territories (Australia, Brazil, Canada, Chile, Mexico, New Zealand, Philippines, South Korea, Taiwan, and Thailand) covering country-specific periods within 2000–2019. All available death records from participating locations were included. Each record included details on age at death, sex, date of death, and the primary cause of death. Cause-specific mortality was classified according to the International Classification of Diseases 10th Revision (ICD-10), covering cardiovascular diseases, respiratory diseases, neoplasms, mental disorders, nervous system diseases, renal diseases, infectious diseases, injuries and external causes of death. Additional details on data sources, collection procedures, and ICD-10 code classifications are provided in [Appendix p 3](#).

Meteorological data and population-weighted exposure estimation

We obtained hourly temperature and dew point temperature data from the fifth generation European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis (ERA5), with a spatial resolution of $0.25^\circ \times 0.25^\circ$ (~25 km). Daily mean temperature and dew point temperature were derived by averaging the hourly values for each grid cell, adjusted to the local time zone. Using these values, we calculated daily relative humidity (RH) based on established formulae.¹⁹ Global population data at 0.01° (~1 km) resolution were obtained from LandScan and further calibrated at the national level according to the population size reported by the United Nations.²⁰ To estimate population-weighted daily temperature and RH for each study location, we aggregated the grid-level meteorological values, weighting them according to the population distribution across all grids within each location boundary. Population-weighted daily meteorological data were then matched with daily all-cause and cause-specific death counts for each location.

Other data sources

We obtained gridded gross domestic product (GDP) and population data at a 0.5° (~55 km) resolution from the Global Carbon Project for the years 2000, 2010, and 2020.²¹ To estimate annual values, we performed linear interpolation on the decadal data. For each location, we identified the corresponding grid cell based on its centroid and extracted the respective GDP and population values. Subsequently, we calculated GDP per capita by dividing the GDP by the population for each location. The relative deprivation index (RDI) was obtained from National Aeronautics and Space

Administration's Socioeconomic Data and Applications Center, which characterises the relative levels of multidimensional deprivation and poverty globally at a high resolution (~1 km).²² The RDI values from all grid cells covering each location's boundary were extracted and averaged to determine the location-specific RDI. Data on Koeppen-Geiger climate classification (KGC) were matched for the centroid of each location.²³ Five main climate groups were classified, including A (tropical), B (arid), C (temperate), D (continental), and E (polar).

Statistical analyses

Assessing the relationship between non-optimal temperatures and cause-specific mortality

We applied a two-stage time-series modelling approach to examine the impact of non-optimum temperatures on cause-specific mortality.⁴ In the first stage, we applied a quasi-Poisson generalised linear regression model for each study location to obtain location-specific estimates. Population-weighted daily temperature was modelled by a distributed lag non-linear model (DLNM) featuring exposure-response and lag-response relationships.²⁴ A quadratic B-spline was applied to the exposure-response curve with three internal knots placed at the 10th, 75th, and 90th percentiles of location-specific temperature distributions.⁴ The lag-response association was modelled with a natural cubic B-spline with an intercept and three internal knots spaced evenly on the log scale. A 21-day lag period was chosen to accommodate the prolonged impact of cold temperatures. To account for potential confounders, we adjusted for RH using a natural cubic spline with three degrees of freedom (dfs), as well as for long-term trends and seasonality using a natural cubic spline of time with eight dfs per year. An indicator for the day of the week was also included in the model. Model selection was informed by prior studies and was validated through model fit comparisons in the sensitivity analyses.^{4,24,25}

In the second stage, we applied a random-effects meta-analytical approach to synthesise location-specific estimates and derive overall associations between non-optimal temperatures and cause-specific mortality. Based on prior research, we considered the following location-level predictors in the meta-regression model: average temperature, temperature range, GDP per capita, RDI, latitude, longitude, country/territory, and climate classification. All combinations of candidate predictors were tested, and the model performance was assessed using the I-square statistics (Appendix pp 3 and 5). Exposure-response curves for locations were further calibrated using the best linear unbiased predictions (BLUPs), leveraging information from locations with similar predictor characteristics.^{25,26} For locations with few deaths where direct estimation was not feasible, the fitted meta-regression model

incorporating location-specific predictors was used to estimate the exposure-response relationships. The minimum mortality temperature (MMT), defined as the temperature with the lowest mortality risk, was derived for each location using the exposure-response curves predicted by the BLUPs. The exposure-response curve for each location was then re-centred using the location-specific MMT as the reference to calculate the relative risk (RR) for a given temperature value across the 21-day lag period. To characterise extreme temperature effects, we estimated the cumulative RR at the 5th (extreme cold) and 95th (extreme heat) percentiles of the location-specific daily temperature distribution relative to the MMT. These percentiles were derived from the full study-period temperature distribution for each location and therefore represent relative extremes within each local climate, rather than uniform absolute temperature thresholds applied across regions. To characterise the demographic profiles of temperature-mortality associations, stratified analyses were performed for each sex and age group (0–59 and ≥ 60). In addition, post hoc analyses were performed for all-cause mortality across four subgroups (men aged 0–59 years, men aged ≥ 60 years, women aged 0–59 years, and women aged ≥ 60 years).

Attributable burden

For each cause of death, we calculated the attributable number of deaths (AD) due to non-optimum temperatures and four components (extreme cold, moderate cold, moderate heat, and extreme heat) according to the method previously described, based on location-specific RR derived from BLUPs, daily series of temperature, and the MMT.^{25,27} Monte Carlo simulations (1000 iterations) were used to obtain the empirical confidence intervals (eCIs).^{28–30} Four components of temperatures were defined as follows: extreme cold (<5th percentile), moderate cold (from 5th percentile to the MMT), moderate heat (from MMT to 95th percentile), and extreme heat (>95th percentile). The attributable fraction (AF) was computed by dividing the AD by the total death counts.

To quantify the premature mortality burden associated with non-optimum temperatures, we estimated temperature-attributable years of life lost (YLL). As a post hoc analysis, for each location, we calculated YLL by multiplying the estimated number of deaths by the standard life expectancy for the corresponding age group, as defined by the Global Burden of Disease 2023 age-specific life tables. The attributable YLL were then calculated following the same method used for the calculation of AD. To account for statistical uncertainty, 95% eCI were generated via Monte Carlo simulations (1000 iterations).

Sensitivity analyses

We performed several sensitivity analyses to test the robustness of our results: 1) we used alternative

maximum lag days ranging from 19 to 23 days; 2) we adjusted the placement of knots in the exposure-response function of temperature and modified the number of knots in the lag-response function of temperature; 3) we changed the df for the long-term trend and seasonality from six to ten per year; 4) as post hoc analyses, we additionally adjusted the first-stage models for daily fine particulate matter (PM_{2.5}) and ozone (O₃) concentrations, incorporating these as covariates to assess potential confounding³¹; 5) as post hoc analyses, we replaced daily mean temperature with apparent temperature (AT) and wet-bulb globe temperature (WBGT) to evaluate sensitivity to the temperature metric.³² We tested the differences between subgroups and between effect estimations from alternative models and our primary models using the fixed-effect meta-regression models.^{33,34} All analyses were performed in R software (version 4.0.3). The “dlnm” and “mixmeta” packages were used to construct the cross-basis function and the meta-analyses.

Ethics

The study was approved by Monash University Human Research Ethics Committee (approval number 29141). The requirement for written informed consent was waived by the ethics committee because the study involved secondary analyses of de-identified data.

Role of the funding source

The funders had no involvement in study design, data collection, data analyses, data interpretation, or the writing of the report.

Results

Descriptive statistics

A total of 46 million deaths were recorded. Cardiovascular causes accounted for the largest proportion of all-cause mortality (26.3%), followed by neoplasms (19.2%) and injury/external causes (10.1%) (Appendix p 6). New Zealand and the Philippines reported a higher proportion of cardiovascular deaths (35.2% and 34.8%, respectively), while Australia (5.8% vs. 1.5% on average) and Canada (4.7%) had a higher share of mortality from mental disorders. Infectious disease mortality was more prevalent in Thailand (9.7% vs. 4.1% on average) and the Philippines (9.3%). Conversely, the Philippines had a low proportion of injury/external cause deaths (5.9% vs. 10.1% on average). The average daily mean temperature across all locations was 21.1 °C (25th–75th: 13.0°C–25.2 °C), ranging from 5.5 °C (–3.0°C–14.7 °C) in Canada to 26.6 °C (25.3°C–27.9 °C) in Thailand. The location-specific temperatures and mortality rates are shown in Fig. 1.

Exposure-response curves

Three distinct exposure-response curve patterns were observed, as shown in Fig. 2. Most cause-specific

mortality had greater risks at extreme cold temperatures than extreme heat (an inverse J-shaped curve). In contrast, mortality from all-cause, respiratory, and nervous system diseases followed a U-shaped curve; and infectious diseases, injury and external cause of death showed a J-shaped curve, with greater risks at extreme heat than extreme cold. Cause-specific exposure-response curves for each country or territory are presented in Appendix pp 7–16. The magnitude of the associations was generally comparable between sexes, although males tended to exhibit slightly higher risks at lower temperatures but lower risks at higher temperatures. Clearer differences were observed across age groups. Individuals aged ≥60 years showed substantially steeper exposure-response curves than those aged 0–59 years, particularly at colder temperatures. Age- and sex-specific exposure-response curves for each cause of death are shown in Appendix pp 17–25. Individuals aged ≥60 years exhibited substantially steeper exposure-response curves for both sexes compared with those aged 0–59 years. Among individuals aged 0–59 years, differences between males and females were small. In contrast, within the older age group, cold-related risks were higher among males than females, whereas heat-related risks were more pronounced among females (Appendix p 26).

Table 1 presents the MMT, percentiles of MMT (MMP), and RR at the extreme cold and heat temperatures for each cause and subgroup. For all-cause mortality, the MMT was 18.6 °C, corresponding to the 57th percentile of temperature distribution. The MMTs varied substantially across different causes of death, ranging from 10.3 °C for injury and external causes to 22.3 °C for mental disorders, with corresponding MMPs ranging from the 11th to the 81st percentiles. The RR of all-cause mortality was 1.038 (95% CI: 1.033, 1.042) under extreme cold and 1.032 (95% CI: 1.027, 1.037) under extreme heat. Mental disorders had the largest RR for extreme cold temperature (RR: 1.095, 95% CI: 1.034, 1.161), followed by renal disease (RR: 1.079, 95% CI: 1.037, 1.123) and cardiovascular diseases (RR: 1.074, 95% CI: 1.062, 1.086). Under extreme heat, the highest mortality risk was observed for nervous system disease (RR: 1.118, 95% CI: 1.079, 1.158), followed by infectious disease (RR: 1.080, 95% CI: 1.053, 1.107) and respiratory disease (RR: 1.074, 95% CI: 1.054, 1.093).

Compared to females, males exhibited higher MMTs for all-cause and most cause-specific mortalities, except for neoplasms, mental disorders, injury and external causes of death. Males generally had higher cold-related mortality risks, while females had greater heat-related risks. Significantly higher RRs under extreme cold temperatures were observed in males compared with females for all-cause, cardiovascular, respiratory, and digestive diseases, whereas females showed significantly higher RRs under extreme heat for all-cause and

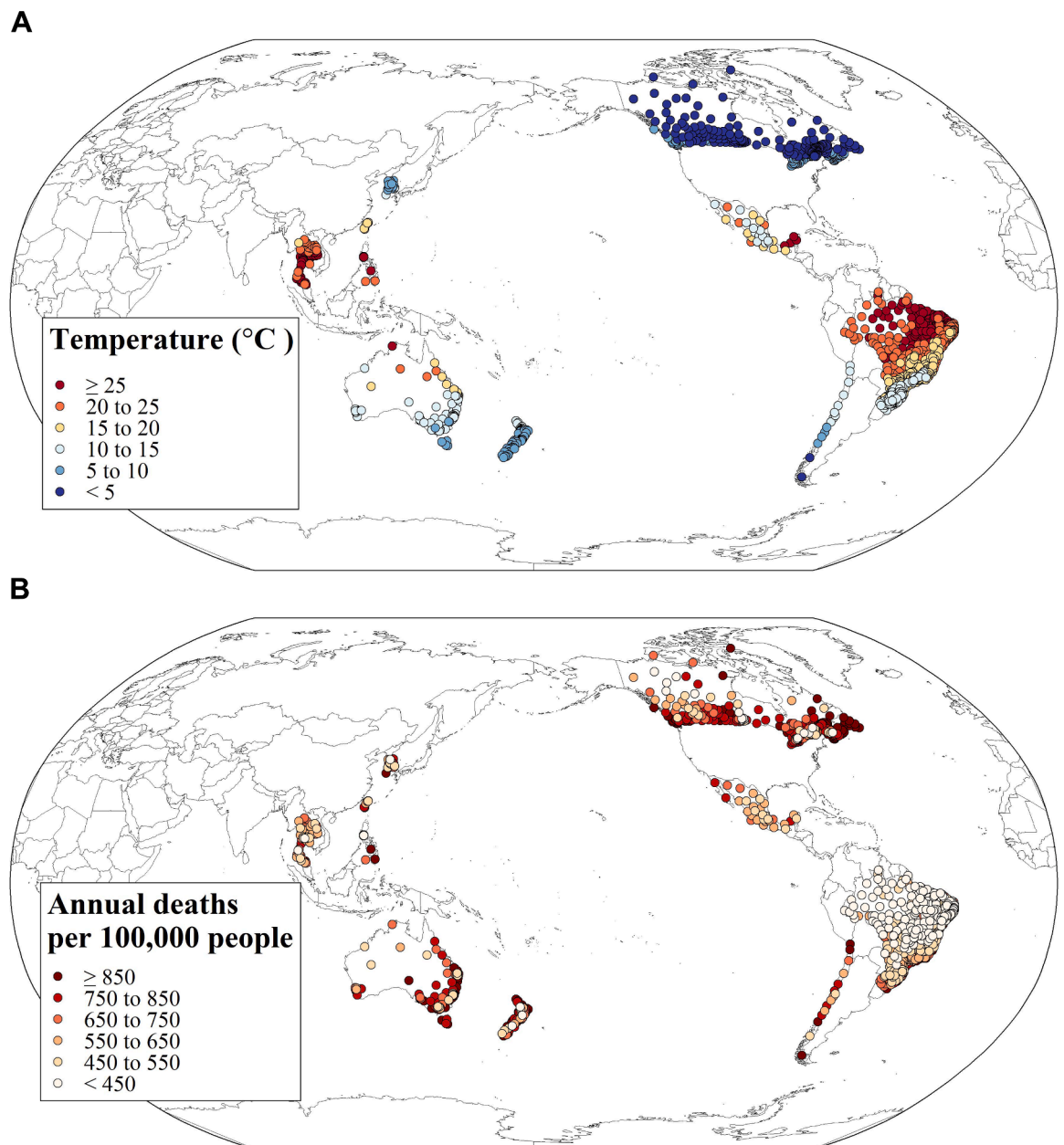


Fig. 1: Annual mean temperature (A) and annual death counts per 100 000 people (B) during the location-specific period in 1117 locations across ten countries or territories.

cardiovascular mortality (all p for difference < 0.05). Those who died aged ≥ 60 years had higher MMTs for all-cause, neoplasms, nervous, infectious, and injury and external causes compared to younger counterparts. Death in people aged ≥ 60 years had higher risks regarding cold exposure compared with younger individuals in all causes except for respiratory mortality, while those who died aged 0–59 years had a

significantly higher RR under extreme heat for injury and external causes of death ($p < 0.001$).

Lag-response curves

At extreme heat temperatures (24.7 °C on average), mortality risk peaked on lag day 0, with the highest risks observed for nervous system disease and mental disorders. The risk then declined to zero by lag day 2–5,

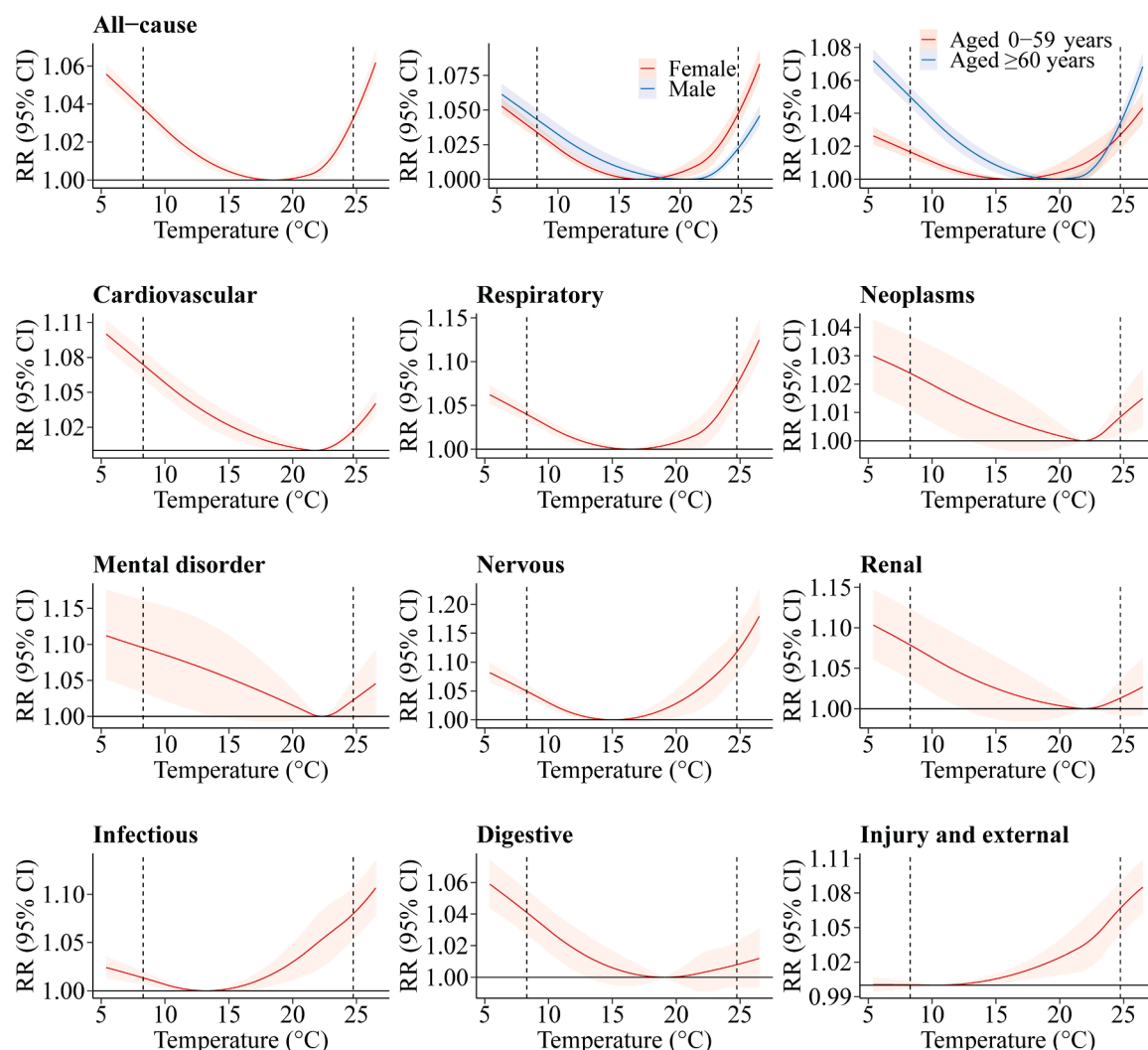


Fig. 2: Cause-specific exposure-response associations between temperature and mortality. Solid lines represent mean estimates; shaded areas indicate 95% confidence intervals. Dashed grey lines are the 5th and 95th percentiles. Abbreviations: RR, relative risk.

depending on the cause of death, followed by a mortality displacement in subsequent days. In contrast, at extreme cold temperatures (8.3 °C on average), mortality risks peaked on lag day 3–5 and gradually declined until lag day 21 (Fig. 3). In particular, cold-related mortality risks of neoplasms demonstrated a shorter lag period of up to ten days.

Attributable burden

A total of 2.03 million deaths (95% eCI: 0.93, 3.05) were attributable to non-optimum temperatures across ten countries or territories over the study period, corresponding to 620 YLL (95% eCI: 281, 936) per 100 000 population (Appendix pp 27–35). Cold temperatures contributed to 1.55 million deaths (95% eCI: 0.64, 2.41), equivalent to 463 years of life lost (95% eCI: 189, 720)

per 100 000 population. Heat accounted for 0.47 million deaths (95% eCI: 0.06, 0.85), corresponding to 157 years of life lost (95% eCI: 19, 285) per 100 000 population. The three leading causes of temperature-related deaths were cardiovascular disease (0.69 million, 95% eCI: 0.63, 0.76), respiratory disease (0.22 million, 95% eCI: 0.19, 0.25), and neoplasms (0.15 million, 95% eCI: 0.10, 0.21).

The fraction of all-cause mortality attributable to non-optimum temperatures was 4.38% (95% eCI: 2.02, 6.59). The highest AF was observed in Chile (AF: 6.25%, 95% eCI: 2.61, 9.60), followed by Canada (AF: 6.00%, 95% eCI: 2.60, 9.12) and Australia (AF: 5.75%, 95% eCI: 1.43, 9.69). Across different causes of death, the highest AF was recorded for mental disorders (AF: 6.53%, 95% eCI: 3.70, 9.23), nervous system disease

	MMT (°C)	MMP (%)	Relative risks (95% confidence interval)			
			Extreme cold (5th percentile)	<i>P</i> _{difference}	Extreme heat (95th percentile)	<i>P</i> _{difference}
All-cause	18.6	57	1.038 (1.033, 1.042)		1.032 (1.027, 1.037)	
Male	20.4	68	1.043 (1.036, 1.050)	0.028	1.022 (1.018, 1.027)	<0.001
Female	16.9	47	1.034 (1.030, 1.038)	Ref	1.048 (1.039, 1.056)	Ref
Aged 0–59	16.1	42	1.017 (1.013, 1.020)	<0.001	1.027 (1.019, 1.035)	0.174
Aged ≥60	19.7	64	1.050 (1.044, 1.056)	Ref	1.034 (1.028, 1.040)	Ref
Cardiovascular	21.7	77	1.074 (1.062, 1.086)		1.017 (1.010, 1.024)	
Male	22.3	81	1.085 (1.068, 1.101)	<0.001	1.012 (1.004, 1.020)	0.036
Female	18.6	57	1.047 (1.038, 1.056)	Ref	1.027 (1.016, 1.037)	Ref
Aged 0–59	22.0	79	1.068 (1.044, 1.093)	0.588	1.016 (1.002, 1.031)	0.995
Aged ≥60	21.7	77	1.076 (1.063, 1.089)	Ref	1.016 (1.009, 1.023)	Ref
Respiratory	16.4	44	1.040 (1.032, 1.048)		1.074 (1.054, 1.093)	
Male	16.4	44	1.038 (1.029, 1.046)	0.016	1.060 (1.038, 1.082)	0.052
Female	15.1	36	1.024 (1.016, 1.031)	Ref	1.093 (1.068, 1.118)	Ref
Aged 0–59	22.1	80	1.125 (1.033, 1.225)	0.082	1.054 (1.025, 1.085)	0.175
Aged ≥60	16.4	44	1.043 (1.035, 1.051)	Ref	1.080 (1.059, 1.101)	Ref
Neoplasms	21.8	78	1.024 (1.011, 1.037)		1.008 (1.002, 1.015)	
Male	19.1	60	1.017 (1.008, 1.027)	0.950	1.004 (0.994, 1.014)	0.787
Female	21.8	78	1.017 (0.998, 1.035)	Ref	1.005 (0.996, 1.015)	Ref
Aged 0–59	16.1	42	1.014 (1.006, 1.022)	0.054	1.006 (0.986, 1.026)	0.524
Aged ≥60	22.0	79	1.031 (1.016, 1.047)	Ref	1.013 (1.005, 1.020)	Ref
Mental disorder	22.3	81	1.095 (1.034, 1.161)		1.023 (0.995, 1.051)	
Male	19.1	60	1.038 (1.013, 1.065)	0.839	1.006 (0.979, 1.034)	0.155
Female	22.0	79	1.033 (0.986, 1.082)	Ref	1.033 (1.008, 1.059)	Ref
Aged 0–59	NA	NA	NA		NA	
Aged ≥60	22.5	83	1.120 (1.039, 1.208)		1.024 (0.992, 1.056)	
Nervous	15.0	35	1.050 (1.039, 1.061)		1.118 (1.079, 1.158)	
Male	15.5	38	1.034 (1.021, 1.047)	0.487	1.060 (1.024, 1.098)	0.195
Female	14.2	30	1.028 (1.017, 1.040)	Ref	1.097 (1.056, 1.139)	Ref
Aged 0–59	13.5	26	1.035 (1.017, 1.053)	0.052	1.139 (1.066, 1.217)	0.409
Aged ≥60	15.9	41	1.059 (1.043, 1.075)	Ref	1.102 (1.059, 1.147)	Ref
Renal	22.0	79	1.079 (1.037, 1.123)		1.013 (0.991, 1.035)	
Male	22.8	85	1.051 (1.006, 1.097)	0.217	1.001 (0.982, 1.021)	0.323
Female	14.8	34	1.021 (1.007, 1.035)	Ref	1.024 (0.984, 1.066)	Ref
Aged 0–59	NA	NA	NA		NA	
Aged ≥60	19.9	65	1.071 (1.038, 1.104)		1.011 (0.987, 1.037)	
Infectious	13.1	24	1.013 (1.007, 1.020)		1.080 (1.053, 1.107)	
Male	14.0	29	1.009 (1.001, 1.018)	0.969	1.039 (1.009, 1.069)	0.2
Female	12.8	22	1.009 (1.000, 1.018)	Ref	1.069 (1.034, 1.106)	Ref
Aged 0–59	12.6	21	1.007 (0.999, 1.016)	0.073	1.053 (1.017, 1.089)	0.148
Aged ≥60	13.7	27	1.019 (1.009, 1.030)	Ref	1.091 (1.054, 1.130)	Ref
Digestive	19.1	60	1.041 (1.027, 1.055)		1.008 (0.994, 1.022)	
Male	26.5	99	1.051 (1.023, 1.079)	0.024	1.005 (0.996, 1.013)	0.127
Female	15.1	36	1.017 (1.007, 1.027)	Ref	1.028 (0.999, 1.058)	Ref
Aged 0–59	26.5	99	1.043 (1.004, 1.084)	0.802	1.001 (0.989, 1.013)	0.357
Aged ≥60	18.8	58	1.049 (1.032, 1.066)	Ref	1.012 (0.993, 1.031)	Ref
Injury and external	10.3	11	1.000 (0.998, 1.003)		1.067 (1.045, 1.089)	
Male	5.4	1	1.001 (0.997, 1.004)	0.732	1.064 (1.045, 1.083)	0.245
Female	14.7	33	1.003 (0.993, 1.012)	Ref	1.042 (1.011, 1.074)	Ref
Aged 0–59	5.4	1	1.003 (0.999, 1.007)	0.088	1.088 (1.064, 1.113)	<0.001
Aged ≥60	17.9	53	1.018 (1.001, 1.035)	Ref	1.022 (0.997, 1.047)	Ref

Abbreviations: MMT, minimum mortality temperature; MMP, minimum mortality temperature percentiles; *P*_{difference}, *p*-value for differences. Note: Estimations in population aged 0–59 years for mental disorder and renal mortality were not available due to relatively small death counts. "Ref" denotes the reference group used for comparison when calculating *p*-values for differences between subgroups. Bold text indicates statistically significant *p*-values for differences between subgroups (*p* < 0.05).

Table 1: The cause-specific relative risks for extreme cold and heat temperatures, with minimum mortality temperature and corresponding percentile, by sex, age, and cause.

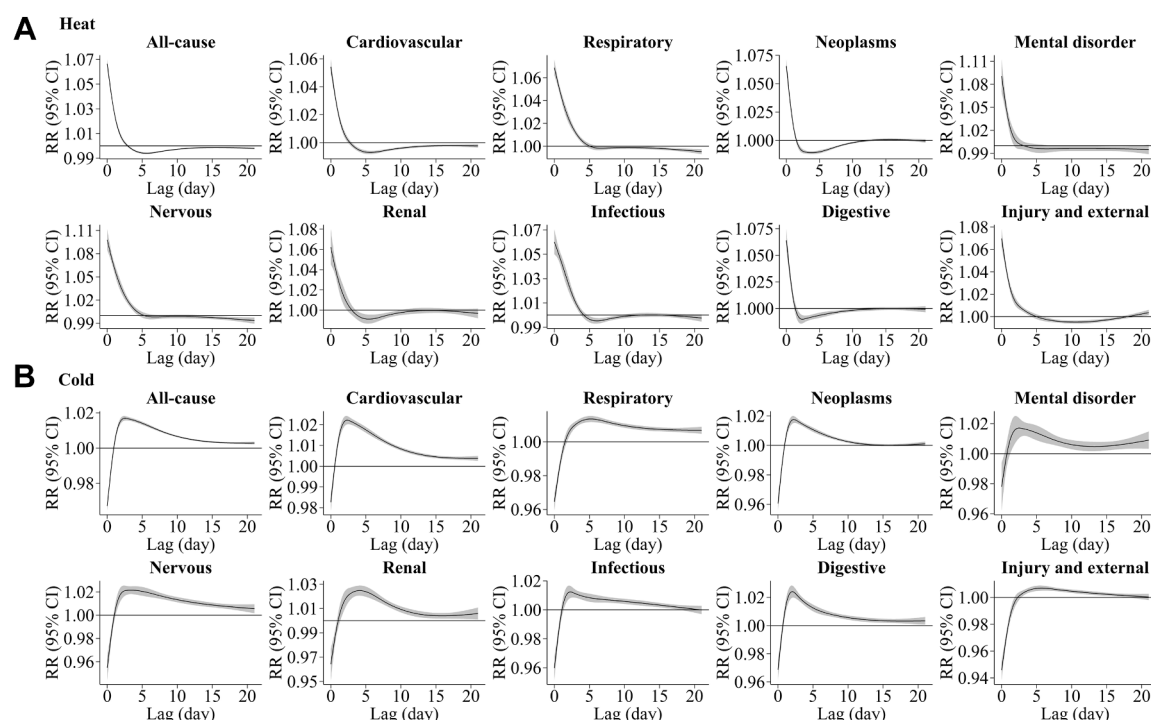


Fig. 3: Lag patterns of the effects of extreme cold (A) and extreme heat (B) on cause-specific mortality. Effects were estimated at 8.3 °C (5th percentile) for extreme cold and 24.7 °C (95th percentile) for extreme heat. Solid lines represent mean estimates; shaded areas indicate 95% confidence intervals. Abbreviations: RR, relative risk.

(AF: 6.40%, 95% eCI: 5.29, 7.48), and cardiovascular disease (AF: 5.71%, 95% eCI: 5.17, 6.25), while the lowest AF was observed for neoplasms (AF: 1.71%, 95% eCI: 1.10, 2.33) (Fig. 4 and Appendix pp 36–39).

For most causes of death, moderate cold contributed the largest share of mortality (AF for all-cause mortality: 2.54%, 95% eCI: 0.53, 4.45), followed by extreme cold (0.82%, 95% eCI: 0.43, 1.16) and extreme heat (0.52%, 95% eCI: 0.19, 0.82). However, for infectious disease, moderate heat had the highest AF (AF: 1.29%, 95% eCI: 0.61, 1.97), followed by extreme heat (AF: 0.70%, 95% eCI: 0.58, 0.82). For injury and external cause of death, most temperature-related mortality was attributed to heat (AF: 3.12%, 95% eCI: 0.85, 5.33), with moderate heat accounting for 2.59% (95% eCI: 0.30, 4.80) and extreme heat for 0.54% (95% eCI: 0.38, 0.69) (Fig. 4 and Appendix pp 36–39).

Sensitivity analysis

A slight increase in the MMT was observed when we changed the maximum lag days from 19 to 23 days, while mortality risks related to extreme cold and heat remained unchanged (all *p* for difference >0.05) (Appendix p 40). No substantial alterations in effect estimates were observed across various model specifications (Appendix pp 41–42). The effect estimates were stable after using different dfs for long-term trend and

seasonality (all *p* for difference >0.05) (Appendix p 43). Additional adjustment for PM_{2.5} and O₃ did not materially alter the effects estimates (all *p* for difference >0.05) (Appendix p 44). Using AT and WBGT slightly reduced the RRs of all-cause mortality at extreme heat from 1.03 to 1.02 (AT) and 1.01 (WBGT), and the RR of cardiovascular mortality from 1.02 to 1.00 (*p* for difference <0.001), while the RRs for other causes remain unchanged (all *p* for difference >0.05) (Appendix p 45).

Discussion

To our knowledge, this is the largest multi-country study to date to characterise cause-, age-, and sex-specific mortality risks associated with non-optimum temperatures. We identified three distinct exposure-response patterns: an inverse J-shaped curve with higher risks from extreme cold for most causes, a U-shaped curve for all-cause and respiratory mortality, and a J-shaped curve for injury and external causes with greater risks at extreme heat. Mortality risk peaked on lag day 0 for extreme heat and lag days 3–5 for extreme cold. Mental disorders had the highest mortality risk under extreme cold, while nervous system disease showed the greatest risk under extreme heat. Critically, differences in temperature-related mortality risks were observed between subgroups of age and sex. We

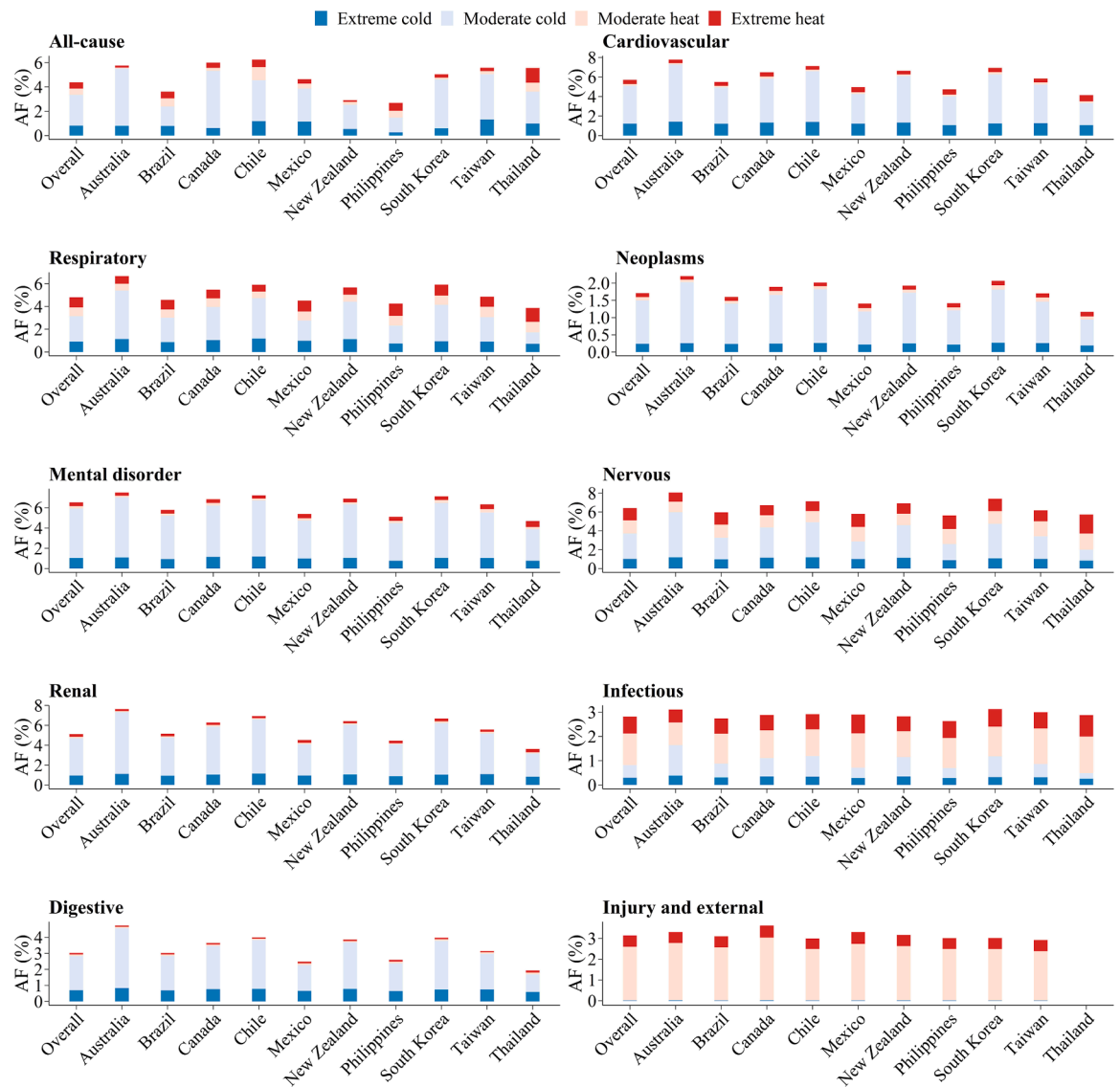


Fig. 4: Cause-specific fraction of mortality attributable to non-optimum temperatures, stratified by country or territory. Attributable fractions are presented as stacked bars, partitioned into contributions from extreme cold (dark blue), moderate cold (light blue), moderate heat (light red), and extreme heat (dark red). Estimates for injury and external causes were unavailable for Thailand due to incomplete cause-of-death data. Abbreviations: AF, attributable fraction.

observed substantial mortality burden attributable to non-optimum temperatures, with the highest AFs estimated for mental disorders, nervous, and cardiovascular causes, and for Chile.

Our findings align with previous studies, which also identified a greater mortality burden from cold than heat.^{1,4} In this study, we extend this evidence by providing cause-specific mortality insights. The inverse J-shaped curves observed for cardiovascular and renal mortality are biologically plausible. Cold exposure induces peripheral vasoconstriction, increases sympathetic activation, elevates blood pressure, enhances

plasma viscosity, and promotes pro-thrombotic states, all of which increase the risk of acute cardiovascular events.^{25,35} Reduced renal perfusion secondary to vasoconstriction and haemodynamic shifts may exacerbate chronic kidney disease and precipitate acute kidney injury.^{36,37} Individuals with limited cardiovascular or renal reserve are therefore particularly vulnerable to cold-induced physiological stress, consistent with the higher cold-attributable fractions observed.³⁸

For mental disorders and nervous system disorders, thermoregulation relies on intact hypothalamic and autonomic function, which may be compromised in

neurodegenerative and psychiatric conditions.³⁹ Autonomic dysregulation can impair vasomotor responses and sweating, reducing capacity to maintain core temperature during environmental extremes. Moreover, several psychotropic medications (e.g., antipsychotics and antidepressants) interfere with thermoregulatory and cardiovascular responses.^{40–42} Among individuals with advanced neurological or psychiatric illness, environmental temperature extremes may therefore precipitate fatal decompensation through autonomic instability, dehydration, and cardiovascular strain. Additionally, cold exposure may increase stress levels (e.g., due to discomfort or worry about debt and affordability), potentially causing psychological changes that impair immune, cardiovascular, and hormonal functions.⁴³

The J-shaped curve observed for injury and external causes, with increased heat-related risks, is consistent with research linking extreme heat to increased accidents, violence, and self-harm.⁴⁴ This association aligns with the temperature-aggression theory, which posits that high temperatures act as environmental stressors that increase aggression and violent behaviour by causing discomfort, sleep deprivation, and heightened sexual arousal.⁴⁵ For infectious diseases, the J-shaped curve indicates a higher heat-related mortality risk, potentially linked to favourable breeding conditions (e.g., rainfall and high humidity levels) during hot seasons, which prolongs vector lifespans and enhances viral replication rates.^{46,47} In addition, heat stress may transiently impair immune function, potentially increasing susceptibility to severe outcomes among infected individuals.⁴⁸ Importantly, although many of the underlying conditions examined are chronic, this does not preclude short-term environmental triggering of death. Within a time-series framework, temperature functions as an acute systemic stressor superimposed on existing vulnerability, increasing the probability of mortality among susceptible individuals rather than inducing disease onset.

The subgroup analysis revealed significant differences in temperature-related cause-specific mortality risks by sex and age groups. Males exhibited significantly higher RRs under extreme cold temperatures for cardiovascular, respiratory, and digestive diseases, whereas females showed significantly higher RRs under extreme heat for cardiovascular disease. These sex-based differences may be partially explained by physiological variations in thermoregulation. Women generally have higher body fat percentages and lower maximal oxygen consumption compared to men, which can influence their ability to dissipate or retain heat under different conditions.⁴⁹ Additionally, differences in skin temperature sensitivity and local cooling responses between sexes further highlight the complexity of thermoregulatory differences.⁵⁰ These physiological factors, combined with social and environmental

determinants, such as housing quality, clothing, and socioeconomic disadvantage, may exacerbate temperature vulnerability, particularly in low-resource settings.⁵¹ Our findings for males are consistent with previous research suggesting that males may be more vulnerable to cold-related health impacts, potentially due to higher prevalence of cardiovascular and respiratory conditions, as well as elevated occupational exposures of middle-aged men.^{52–56} However, this contrasts with evidence from China reporting higher cold-related mortality and morbidity risks among females than males.^{57–59}

Age-specific analyses further indicated that populations aged ≥ 60 years had higher RRs under extreme cold for nervous system diseases and neoplasms, which may reflect physiological and cognitive decline, as well as reduced thermoregulatory capacity associated with ageing and pre-existing conditions.^{57,60,61} In contrast, individuals aged 0–59 years had significantly higher RRs under extreme heat for injury and external causes of death, potentially due to greater outdoor exposure, risk-taking behaviours, and occupational hazards.⁶² These findings highlight the need for tailored public health strategies to address the unique vulnerabilities of specific demographic groups and mitigate temperature-related health risks. For example, older adults (≥ 60 years of age) and individuals with pre-existing conditions require special attention for the exacerbation of neoplasms and nervous system diseases during extreme heat events. In contrast, individuals aged 0–59 years require greater attention for injury and external causes of death associated with heat exposure.

The overall AF of 4.38% observed in this study is within the range reported in previous multi-country analyses but lower than some global modelling estimates. For example, Gasparrini et al. reported country-specific AFs ranging from 3.37% in Thailand to 11.00% in China using a comparable distributed lag framework.⁴ For countries included in both analyses, our estimates were broadly similar, suggesting methodological consistency. In contrast, Zhao et al. estimated a global AF of 9.43%, higher than the present study.¹ Differences may reflect variations in geographic coverage, population structure, climatic distribution, and exposure weighting strategies. Our study includes ten countries/territories and applies population-weighted temperature exposure, which may reduce the influence of sparsely populated regions experiencing extreme temperatures. In addition, differences in study period and cause-specific mortality composition may contribute to variation in overall AFs. These methodological distinctions should be considered when comparing burden estimates across studies.

The findings of this study have important public health and clinical implications. Understanding the distinct temperature–mortality relationships across various causes of death can help guide targeted

interventions. Among the included countries or territories, the attributable mortality burden from cardiovascular, renal, mental disorders, and nervous system diseases was higher than that for other causes of death. Cold temperatures accounted for the largest share of this burden due to the relatively higher proportions of moderate and extreme cold days, emphasising the need for increased caution during cold periods. In contrast, for infectious diseases and injury/external causes of death, greater attention is required during heat events. Healthcare systems could prioritise tailored heat and cold management plans for at-risk groups, especially in regions with higher mortality burdens. For instance, Australia experienced the highest temperature-related mortality burden for most chronic diseases, highlighting the need for improved housing insulation and effective heating strategies to mitigate cold-related risks, as well as implementing urban cooling strategies and heatwave early warning systems to reduce heat-related mortality.^{63–65}

This study has several strengths. Firstly, we used the largest databases covering ten countries or territories, enabling the assessment of temperature-related mortality risks even for relatively rare causes of death. Secondly, the inclusion of 1117 locations across diverse climates and socio-economic conditions enhances the external validity and generalisability of our findings. Thirdly, this study provides valuable insights into how age and sex influence vulnerability to temperature extremes, highlighting specific at-risk populations for different causes of death, which has been rarely investigated in previous multi-country studies on temperature.^{4,14}

However, several limitations must be acknowledged. We relied on grid-level temperature data instead of individual-level measurements, potentially leading to exposure misclassification and underestimation of the true effects.⁶⁶ As with most ecological studies, we could not control for individual-level confounders such as comorbidities, socioeconomic status, or behavioural factors. Although the mortality data are considered nationally representative, the analytical unit (“location”) varied in geographical scale across countries (e.g., cities, communities, provinces), and coverage gaps in some countries or territories may have led to underestimation of attributable deaths. In addition, despite the inclusion of 1117 locations across ten countries/territories, regions such as Africa and the Middle East were not represented, and low-income settings were limited. Differences in climate patterns, infrastructure, healthcare access, and adaptive capacity in these underrepresented regions may influence temperature–mortality relationships. Therefore, caution is warranted when extrapolating these findings to other regions. Despite stringent quality control of the death registry, the potential for diagnostic or coding errors remains, and the quality of death records may vary across the study

period and between countries or territories. Finally, this study does not provide projections of temperature-related mortality under future climate or demographic change scenarios. Producing such projections requires integration of complex modelling frameworks that account for evolving climate conditions, population growth and ageing, and potential shifts in adaptation capacity; each of which introduces substantial uncertainty. Future research should incorporate these dynamics to inform long-term health risk planning under climate change.

In conclusion, this study identified three distinct exposure-response patterns (inverse J-shaped, U-shaped, and J-shaped) with varying MMTs and mortality risks across nine causes of death. Significant differences in temperature-related mortality risks were observed across age and sex groups, with the direction and magnitude of these differences varying by cause of death. Geographical variations highlighted a high mortality burden from non-optimum temperatures in Chile, followed by Canada and Australia. These findings enhance understanding of cause-specific mortality risks linked to non-optimum temperatures and could inform targeted clinical and public health strategies to mitigate the temperature-related health burden.

Contributors

BW, YG, and SL designed the study and statistical methods. BW took the lead in the statistical analyses, visualisation, drafting of the manuscript, and interpretation of the results. YG, SL, and BW accessed and verified the underlying data. All authors contributed to interpreting the results and revising the manuscript, and were responsible for the decision to submit the manuscript for publication.

Data sharing statement

Daily mortality data for 1117 locations are derived from multiple countries. Due to ethical and legal restrictions, these data cannot be made publicly available. Other data are publicly available. Analytical code supporting the findings is available upon reasonable request. Researchers may contact yuming.guo@monash.edu or shanshan.li@monash.edu for information on accessing the analytical codes and restricted daily mortality data.

Editor note

The Lancet Group takes a neutral position with respect to territorial claims in published maps and institutional affiliations.

Declaration of interests

We declare no competing interests.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.eclinm.2026.103920>.

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